

Optimization of Photovoltaic and Battery Storage Systems

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Motivation

2. Overview

3. Requirements

4. Project-Timeline

5. Conceptual Model

6. Assignment

Decentralized energy systems are becoming increasingly important for reducing energy costs and enhancing self-sufficiency. An optimally sized photovoltaic (PV) and battery system enables:

- Minimization of grid electricity purchase costs
- Maximization of on-site energy self-consumption
- Flexible trading of surplus energy with the grid or local networks

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Objective

Optimize the design and operation of photovoltaic and battery storage systems to maximize financial savings.

Method

Simulating every combination of optimization variables within the given constraints in AnyLogic. Allowing a full overview of all possible approaches.

Key Elements

- Selection of PV module count, battery size, and trading strategy
- Integration of user-specific data and real-world constraints
- Consideration of consumption profiles, tariffs, and seasonality

Outcome

An individualized, simulation-driven energy configuration with maximum long-term cost efficiency.

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User Input

- Number of residents, living area
- Electric vehicle ownership and usage
- Heat pump type or coefficient of performance (COP)
- Available budget

Physical Constraints

- Roof area and tilt
- Maximum number of PV modules
- Maximum battery capacity

Internal Variables

- Seasonal effects on PV production and heat demand
- Changing electricity prices
- Varying electricity consumption over the day

Trading Strategies

- Predefined strategy selection for energy buying/selling
- No development of new strategies

1. Motivation

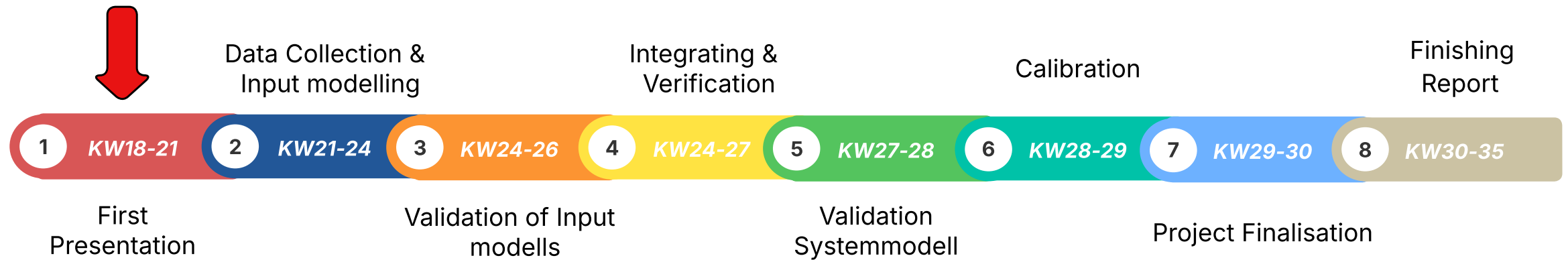
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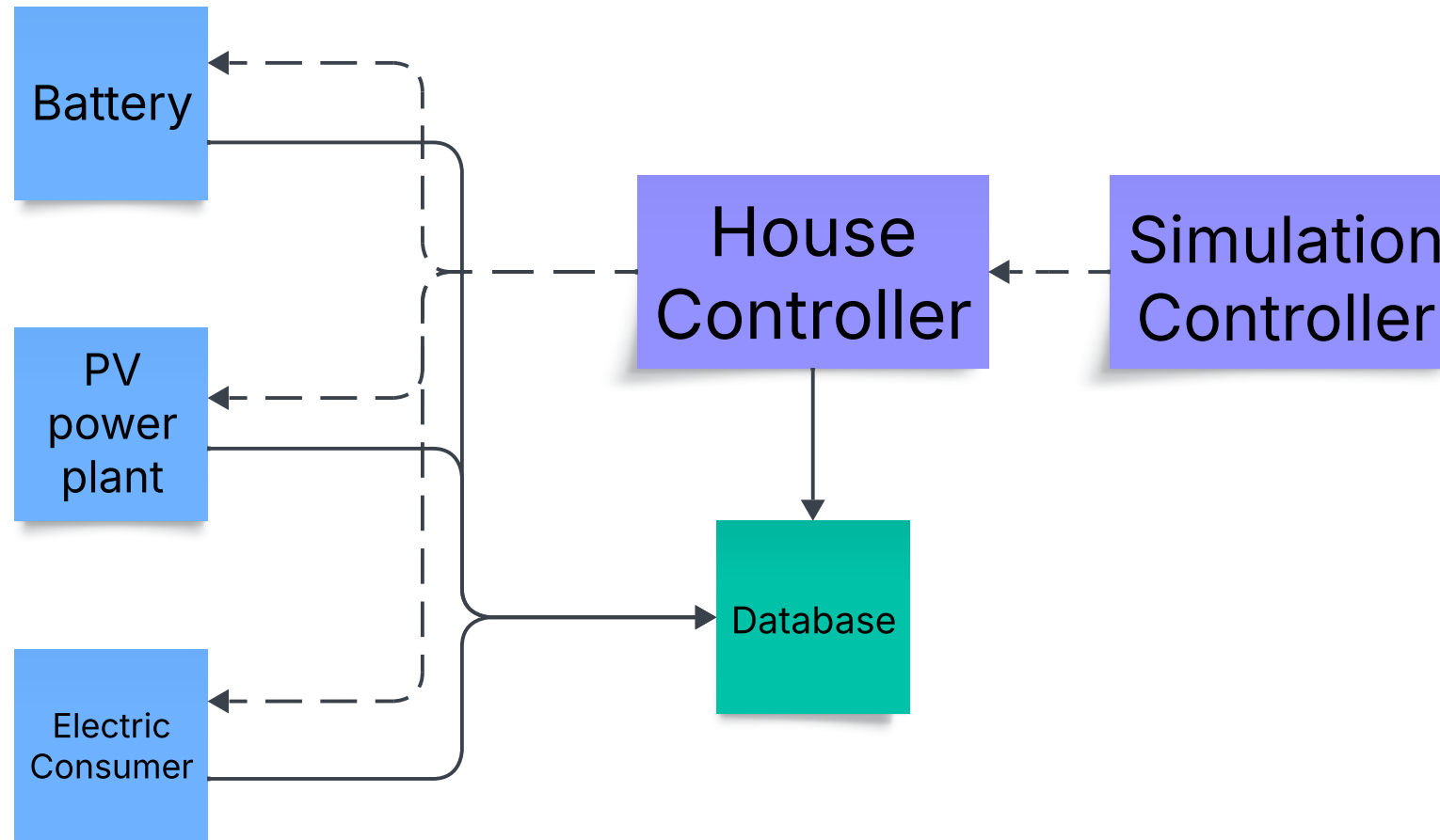
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Conceptional Model



[Click here for detailed Conceptual Model](#)

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Assignment

Assignment



KW18-21	KW21-24	KW24-26	KW24-27	KW27-28	KW28-29	KW29-30	KW30-31
Concept Model Creation Robert Leichtl Florian Merlau Jakob Haberstock Christian Besold	Update & Report Submission Robert Leichtl Florian Merlau Jakob Haberstock Christian Besold Nandini Joshi	Validate Basic System Robert Leichtl Florian Merlau Jakob Haberstock Christian Besold Nandini Joshi					
UML Diagram Robert Leichtl Florian Merlau Jakob Haberstock Christian Besold	Simple Battery Robert Leichtl	Validate Individual Systems Robert Leichtl Florian Merlau Jakob Haberstock Christian Besold Nandini Joshi	Finishing Electric Vehicle Christian Besold Nandini Joshi	Validate Electric Vehicle Florian Merlau Jakob Haberstock			
Division of teams Robert Leichtl Florian Merlau Jakob Haberstock Christian Besold	Simple Person Energy Nandini Joshi	Add Heat Pump Florian Merlau Jakob Haberstock	Finishing up Systems Robert Leichtl Florian Merlau Jakob Haberstock	Validate Heat Pump Christian Besold Nandini Joshi	Calibration and Production Runs Robert Leichtl Jakob Haberstock Christian Besold	Final Presentation Robert Leichtl Florian Merlau Jakob Haberstock Christian Besold Nandini Joshi	Final Report Robert Leichtl Florian Merlau Jakob Haberstock Christian Besold Nandini Joshi
Presentation Robert Leichtl (Presenter) Florian Merlau Jakob Haberstock (Presenter) Christian Besold Nandini Joshi	Simple PV Christian Besold	Extend Systems Robert Leichtl Florian Merlau Jakob Haberstock Christian Besold Nandini Joshi	Showcase Presentation Robert Leichtl Florian Merlau Jakob Haberstock Christian Besold Nandini Joshi	Validate Complete System Robert Leichtl Florian Merlau Jakob Haberstock Christian Besold Nandini Joshi	Animation Florian Merlau Nandini Joshi		
	Simple Database Florian Merlau	Add Simulation Controller Robert Leichtl					
		Simple Electric Vehicle Christian Besold Nandini Joshi					

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